

Class Test 2 (18 August 2026)

1. [100 marks]

Recall the definitions of *minor* and *cofactor*.

Minor of order k : For any $m \times n$ matrix A , consider the k -th order submatrix M obtained by deleting all but some k rows and k columns of A . Then $|M|$ is called a k -th order minor of A .

Cofactor: The cofactor A_{ij} of the element a_{ij} of any square matrix A is $(-1)^{i+j}$ times the determinant of the submatrix obtained from A by deleting row i and column j .

In this question we will prove that if $\text{rank}(A) = k$ then every minor in A of order $k+1$ vanishes, while there is at least one minor of order k that does not vanish.

- Suppose $\text{rank}(A) = k$.
- (a) [30 marks] Argue that the columns of any submatrix of order $k+1$ can be expressed as linear combinations of k columns from A . Prove that all minors of order $k+1$ vanish.
- Now we will show that there is at least one minor of order k that does not vanish. Assume that the opposite holds, that is, that all determinants of order k vanish.
- (b) [54 marks] Select k columns from A which are linearly independent. Without loss of generality, we can consider them to be the first k columns. Assume that all determinants of order k formed from the first k columns vanish.

– [6 marks] Argue that $\sum_{i=1}^k a_{ki} A_{ki} = 0$.

– [15 marks] Consider the $k \times k$ submatrices formed using the first $k-1$ rows and any other row j , $j = k+1, k+2, \dots, m$.

Explain clearly that $\sum_{i=1}^k a_{ji} A_{ki} = 0$, $j = k+1, k+2, \dots, m$.

– [6 marks] Argue that $\sum_{i=1}^k a_{ji} A_{ki} = 0$, $j = 1, 2, \dots, k-1$.

- [18 marks] Using the last few steps argue that the first k columns of A are *linearly dependent* if A_{k1}, A_{k2}, A_{kk} are not all zero.
 - In fact, if there were any minor of order $k - 1$ in the first k columns which did not vanish, we could rearrange the rows and columns so that at least one A_{ki} , $i = 1, 2, \dots, k$, would be different from zero, as in the last step.
 - [9 marks] Summarize, from the steps in this part, what you can conclude about the determinants of order $k - 1$ formed from the first k columns of A starting with the assumption that all determinants of order k formed from the first k columns vanish.
- If all determinants of order $k - 1$ formed from the first k columns of A were zero, we could repeat the procedure with $k - 1$ columns and show that the $k - 1$ columns are linearly dependent if the determinants of order $k - 2$ do not vanish.
- (c) Conceivably, we will in this way arrive at a point where all determinants of order 2 vanish. Now we have reduced the problem to showing that any set of two columns is linearly dependent.
- [16 marks] Argue that this linear dependence would follow.
- Thus, using the above procedure, we have contradicted our assumption that the k columns were linearly independent. Hence, there must be at least one nonvanishing determinant of order k in any set of k linearly independent columns.